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Embedded System Design
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PL resources –Zedboard Artix-7 FPGA technology

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DSP Slices

Feature	Value
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Architecture	DSP48E1
Count	220
Multiplier	25 x 18 bit
Accumulator	48 bit

Operations	

$A \times B$	Multiply
$(A \times B) + C$	Multiply-Accumulate (MAC)
$(A + B) \times C$	Pre-add then multiply
$A + B$	Fast Adder/Accumulator
$A - B$	Subtractor

Supports pipelining– will be discussed later.

Block RAM

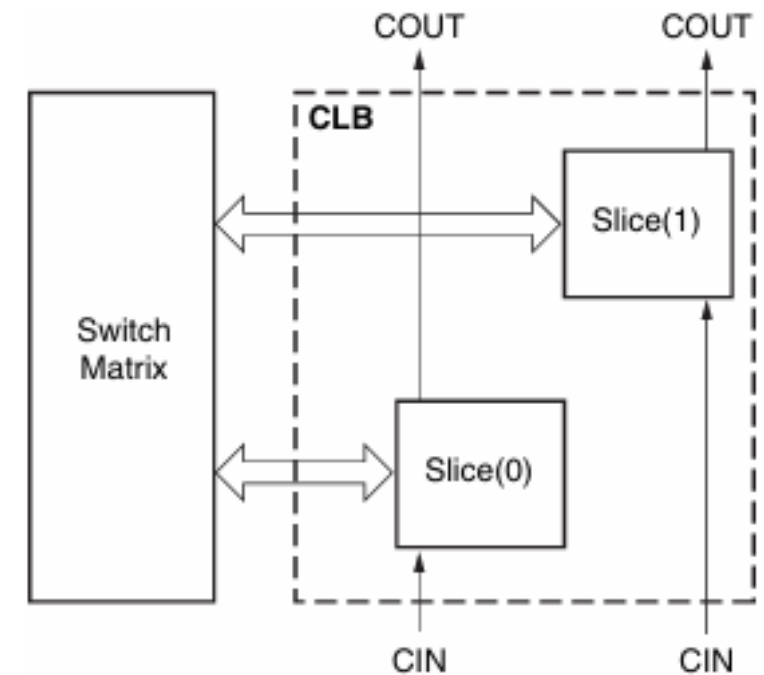
- It is used heavily for **buffers, FIFOs, local data storage, small memories**
- 36kB
- 140 Number
- Total size - $140 \times 36 \text{ Kb} = 5040 \text{ Kb} \approx 630 \text{ KB}$

Configurable Logic Block (CLB)

- A CLB is the fundamental component of an FPGA, allowing the user to implement logical functionality (Sequential or combinational) within the chip.
- In 7-Series FPGA, CLB will have 2 slices.
- Slices are SLICEM, SLICEL
- Each CLB can contain either a SLICEM and SLICEL or two SLICELs.
- There are roughly twice as many SLICELs than SLICEMs on a chip, though the exact numbers vary by device.
- These slices contain four look-up tables (LUTs), eight flip-flops (FF), a network of carry logic, and multiplexers .

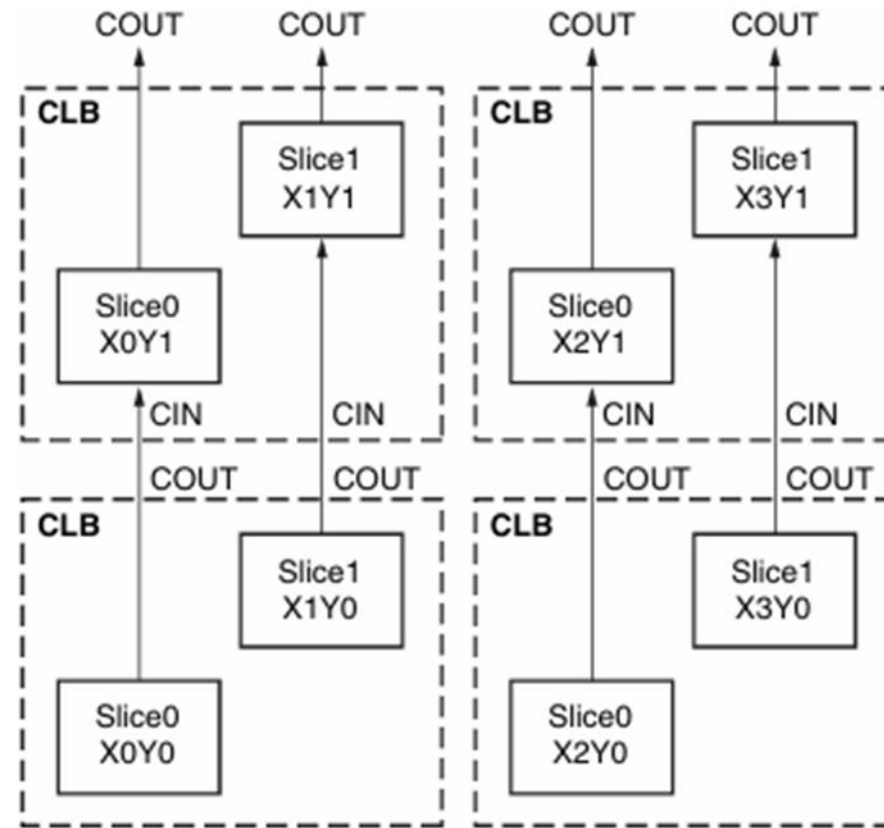
CLB

- A CLB element contains a pair of slices
- Each CLB element is connected to a switch matrix for access to the general routing matrix
- 1 CLB = 2 slices, one slice = 4 LUTs + 8 FF. Therefore, one CLB = 8 LUTs + 16FF.



Arrangement of Slices within the CLB

CLB



Row and Column Relationship between CLBs and Slices

Every Slice contains

- Every slice contains:
 - Four logic-function generators (or look-up tables)
 - Eight storage elements
 - Wide-function Multiplexers
 - Carry logic
- In addition, some slices support two additional functions: storing data using distributed RAM and shifting data with 32-bit registers.
- Slices that support these additional functions are called SLICEM; others are called SLICEL.

Look-Up Table (LUT)

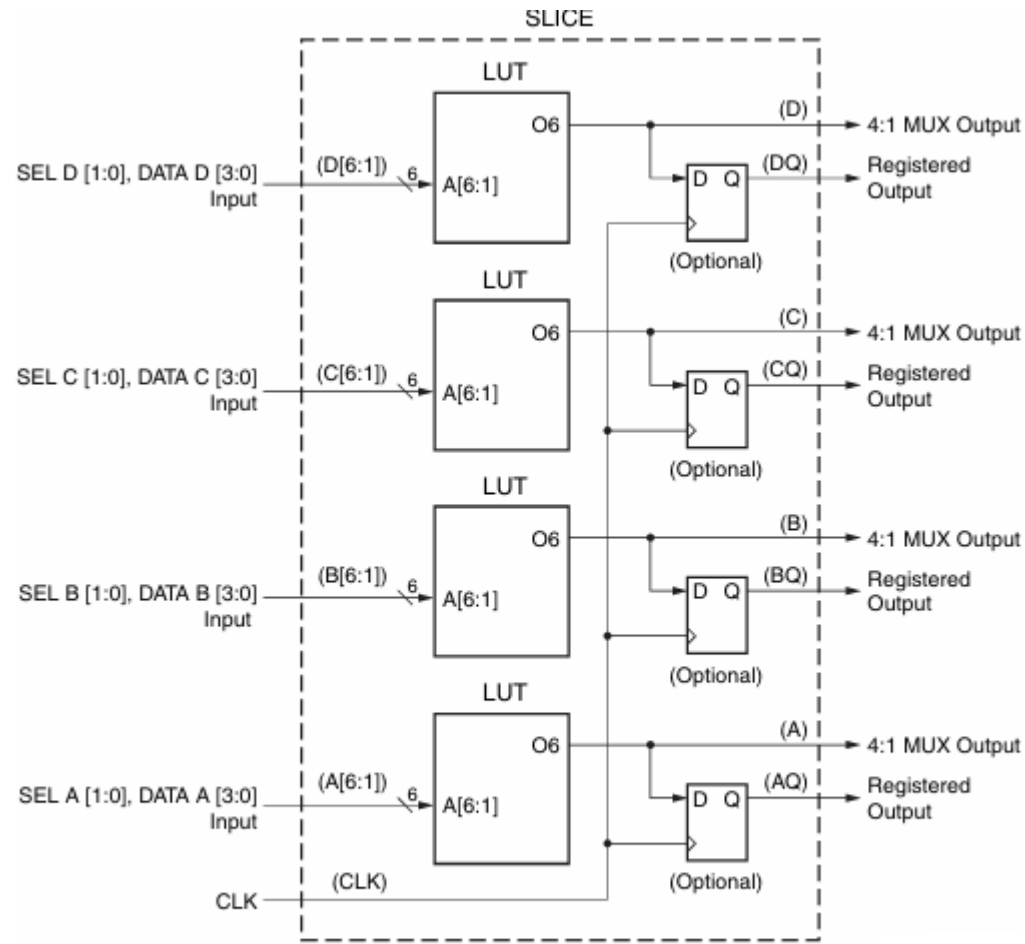
- Slices contain four **look-up tables (LUTs)**, eight flip-flops (FF), a network of carry logic, and multiplexers.
- A six-input function uses:
 - A1-A6 inputs
 - O6 output
- Two five-input (fewer inputs) are used:
 - A1–A5 inputs
 - A6 driven High
 - O5 and O6 outputs

O6 exits slice O5 used internally

Eight storage elements

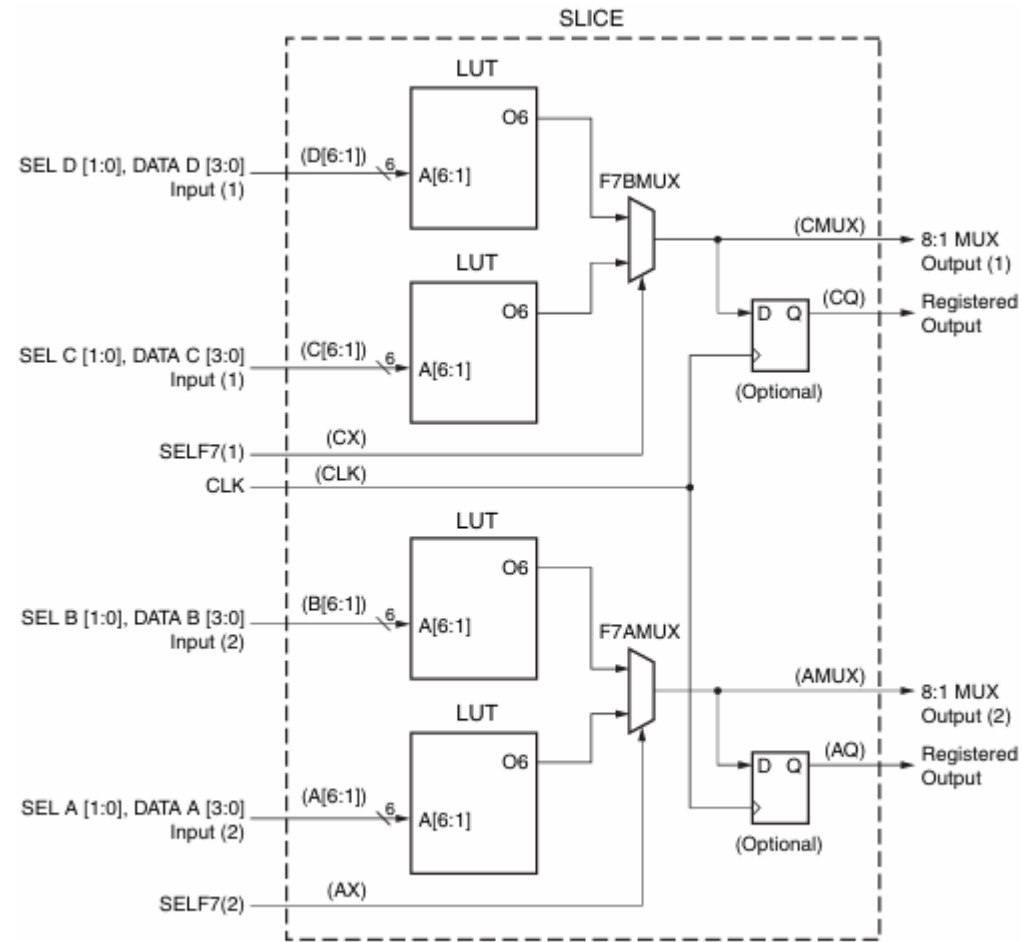
- Each output of the LUT is fed into a storage element
- 8 FF
- 4 latches, 4 unused.

LUT as Multiplexor (4:1)



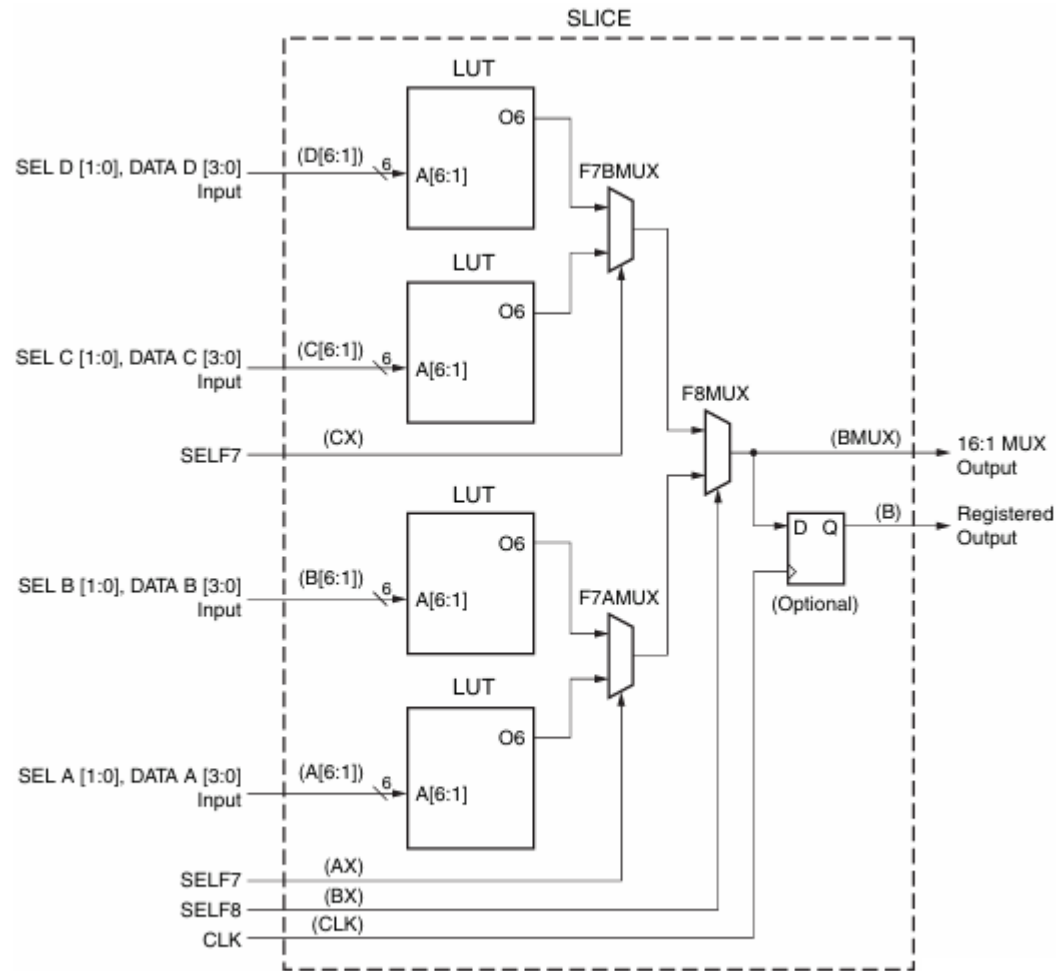
Four 4:1 Multiplexers in a Slice

LUT as multiplexor (8:1)



Two 8:1 Multiplexers in a Slice

LUT as multiplexor (16:1)



16:1 Multiplexer in a Slice

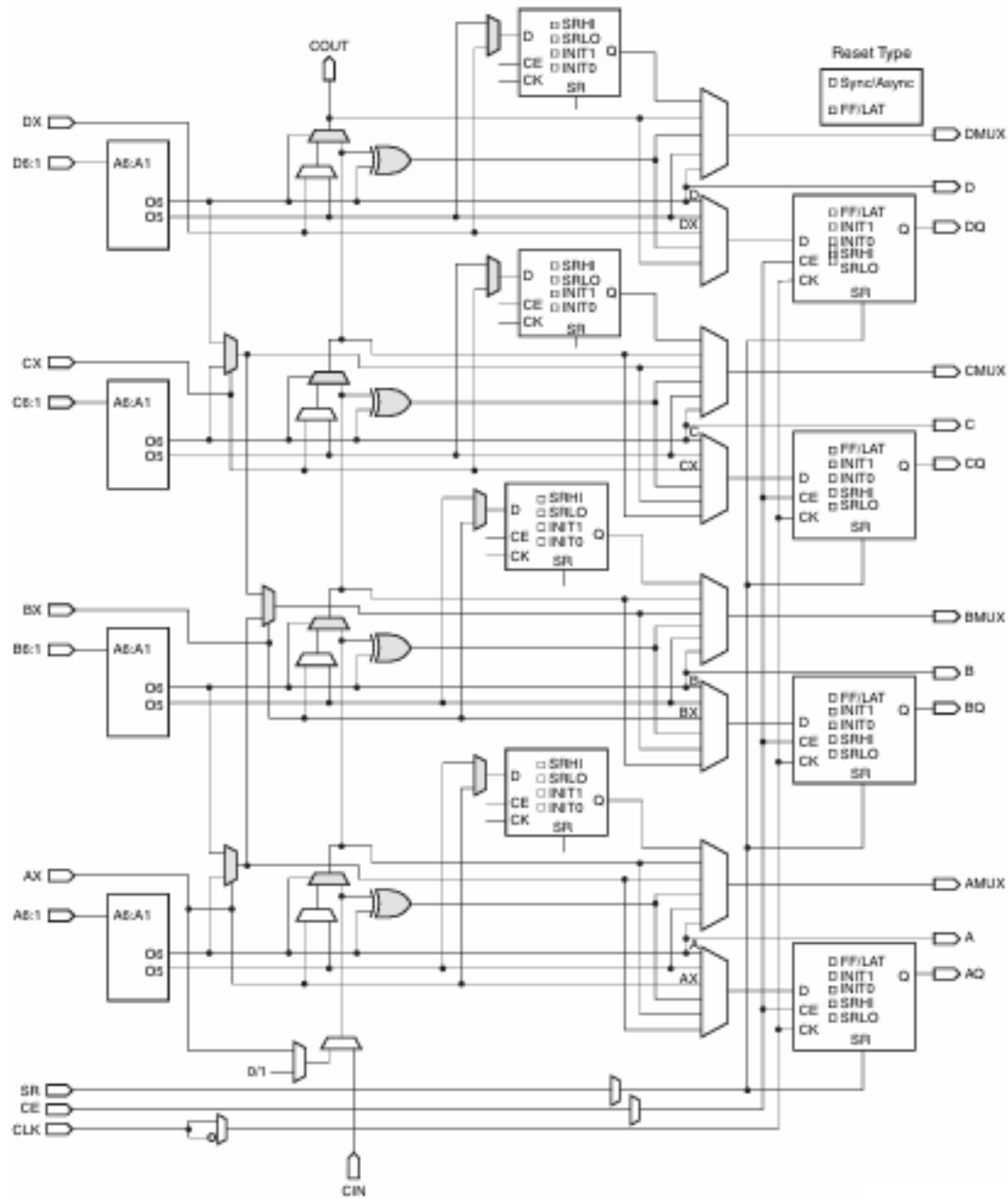


Diagram of SLICEL

- Each input signal (AX, BX, CX, DX) enters an individual LUT (A6:A1, B6:B1, C6:C1, D6:D1). Each LUT can be used independently (6-input), or paired to function as dual 5-input LUTs.
- The output of each LUT (A, B, C, D) is routed directly to either combinational output or sequential logic (flip-flop) via AMUX, BMUX, CMUX, DMUX.
- The primary outputs of a slice are labeled A, B, C, D (combinational outputs) and AQ, BQ, CQ, DQ (registered outputs). These are the signals accessible from outside the slice (i.e., to the FPGA routing fabric).

AX/BX/CX/DX inputs come from the FPGA's general interconnect/routing matrix, just like the regular LUT inputs (A1–A6, B1–B6, etc.). They are independent from the LUT logic signals.

Each of AX, BX, CX, DX is directly connected to one of the four storage elements (flip-flops/latches) in the slice.

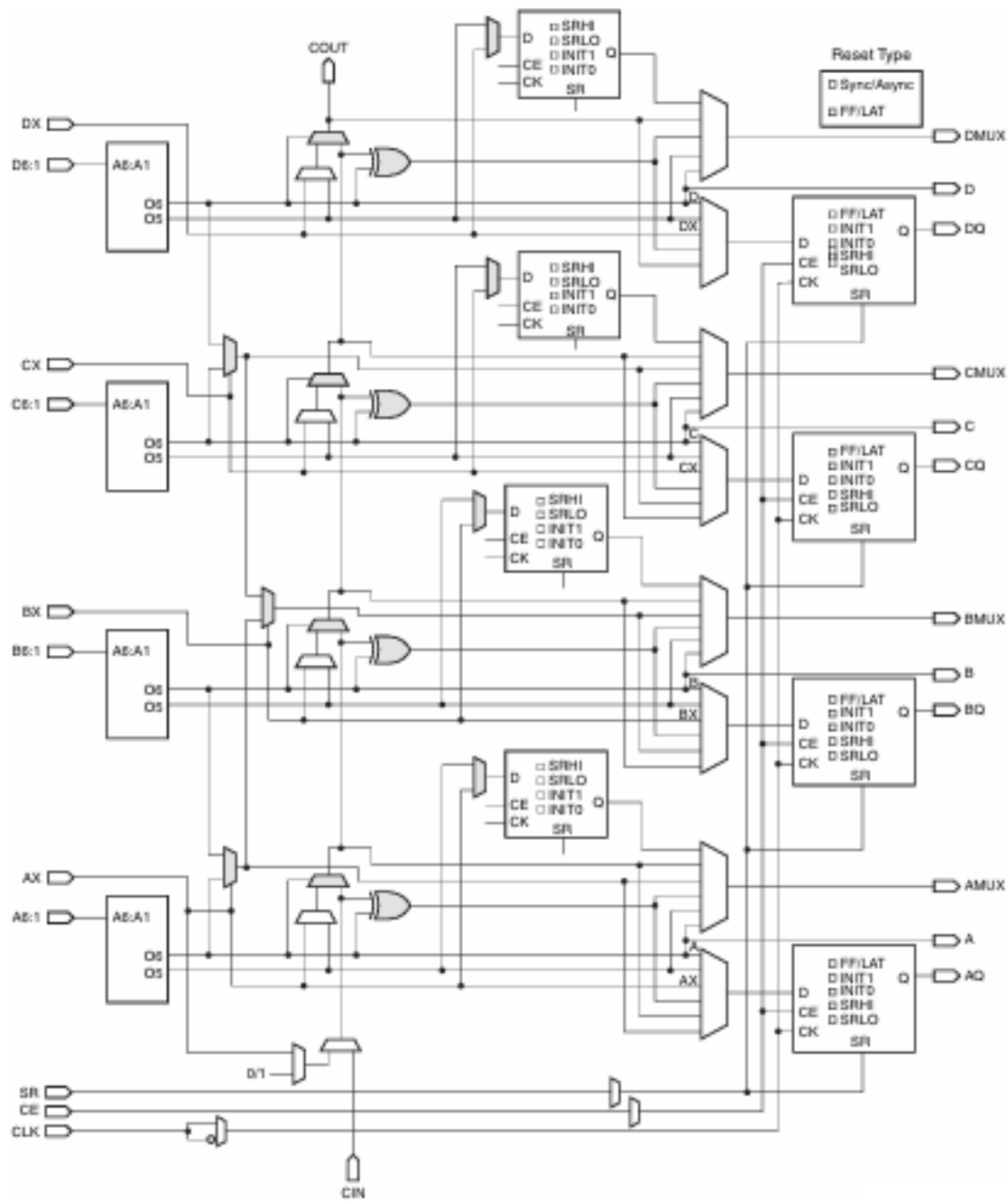


Diagram of SLICEL

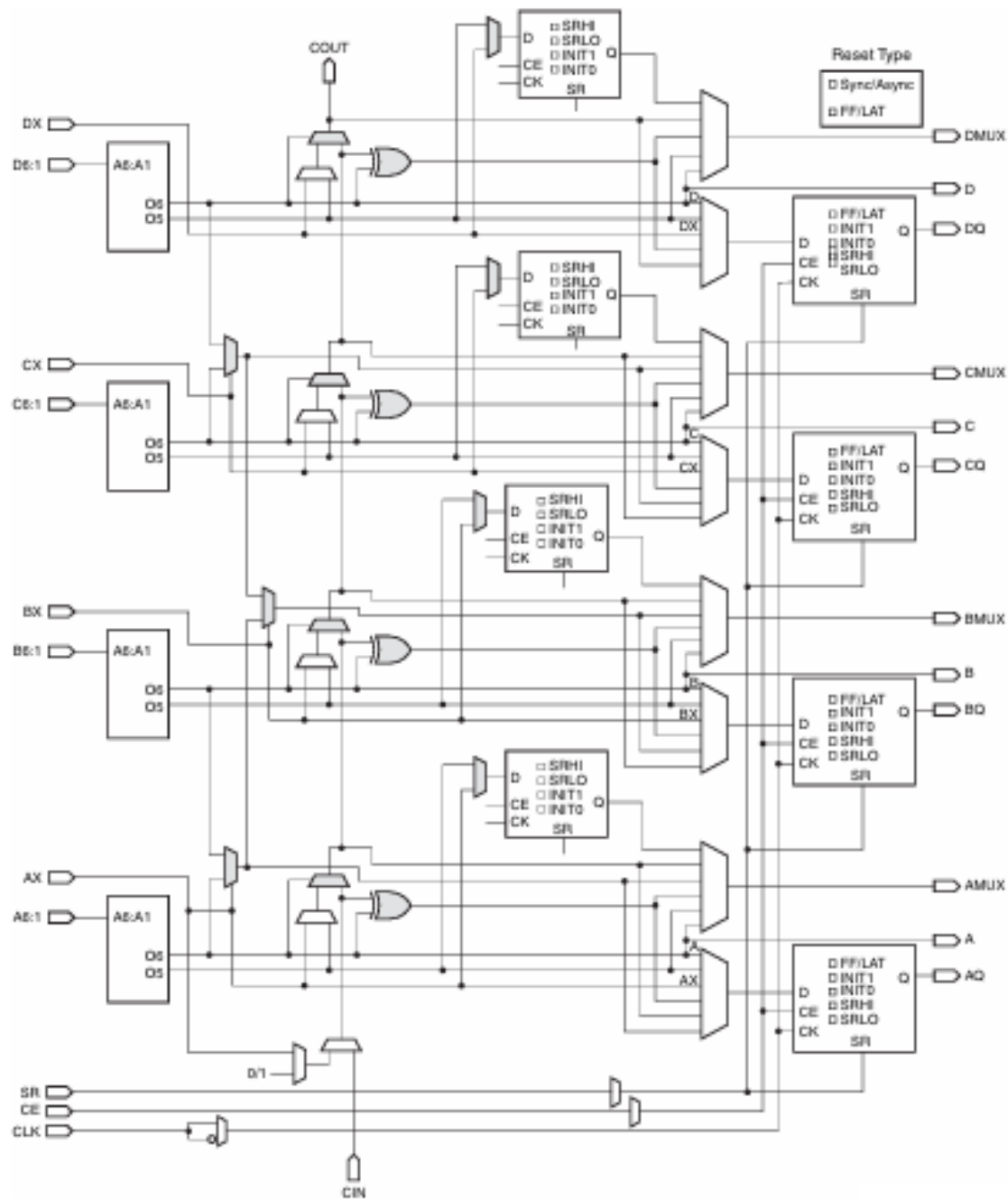


Diagram of SLICEL

- The main output from each LUT (often called O6) is routed to multiple destinations:
 - As input to the carry logic.
 - As an option to the flip-flop (for registering).
 - As an input to the output multiplexer (AMUX, BMUX, etc.).
- XOR gates: Combine inputs to create the sum output.

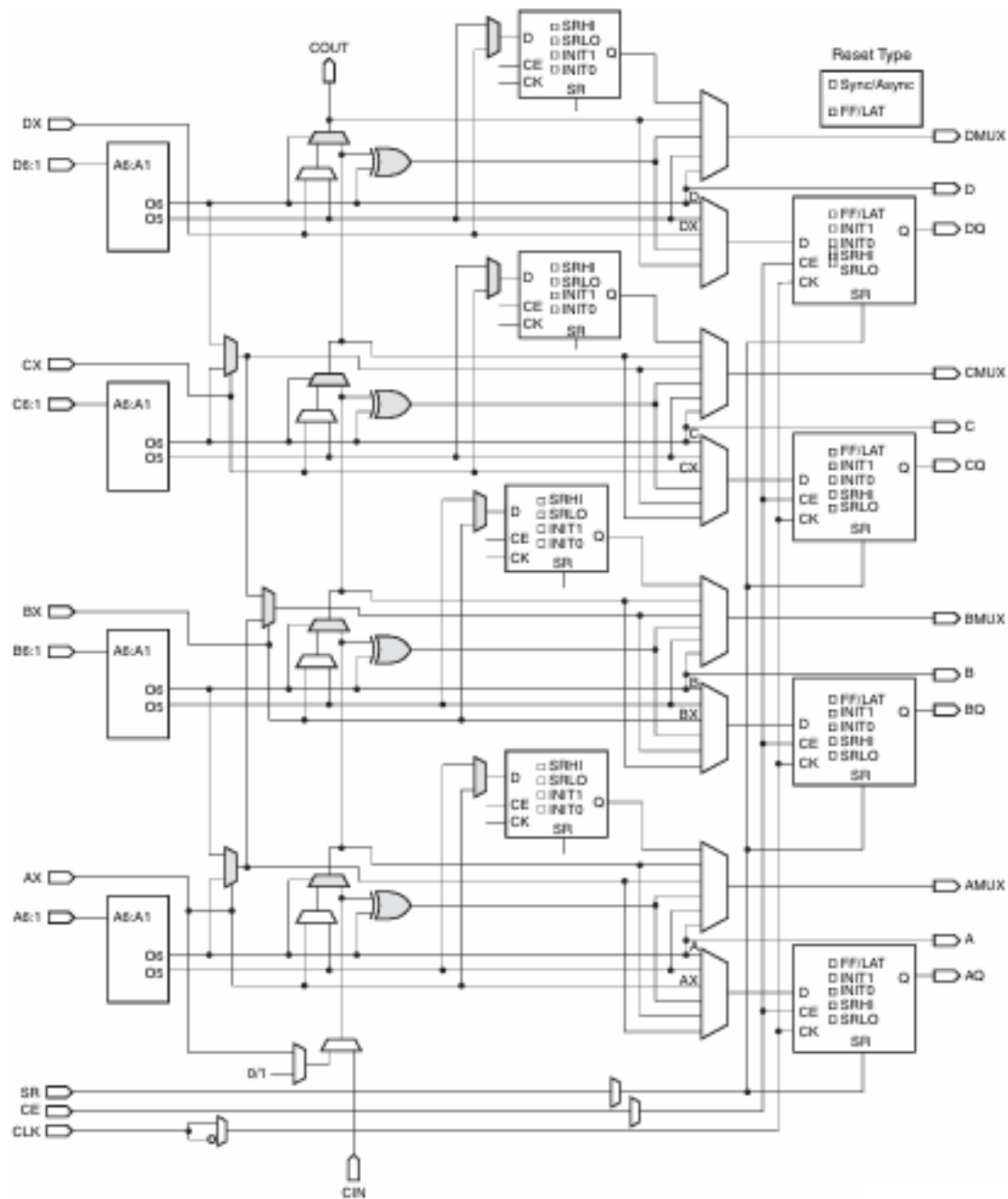
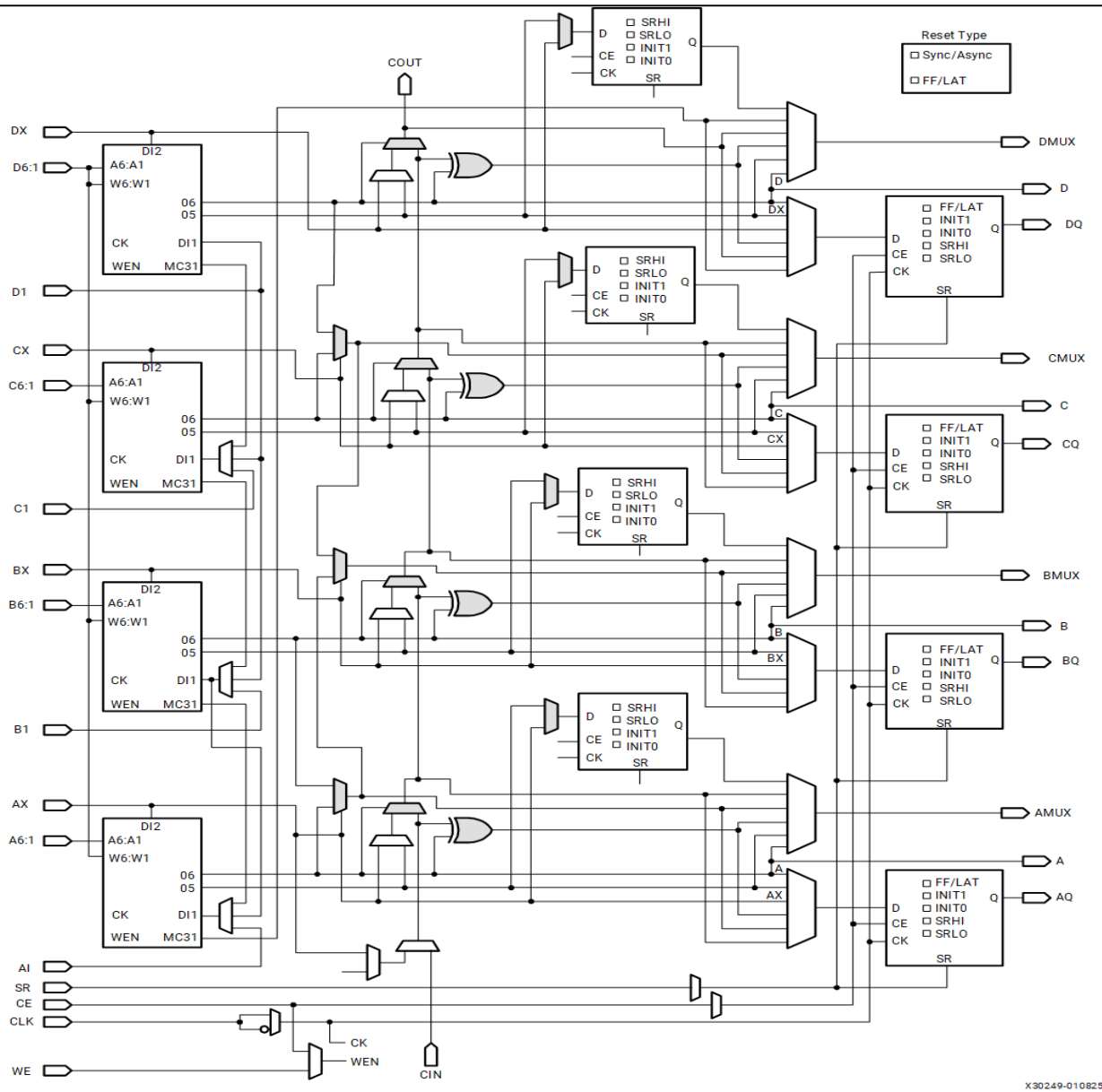


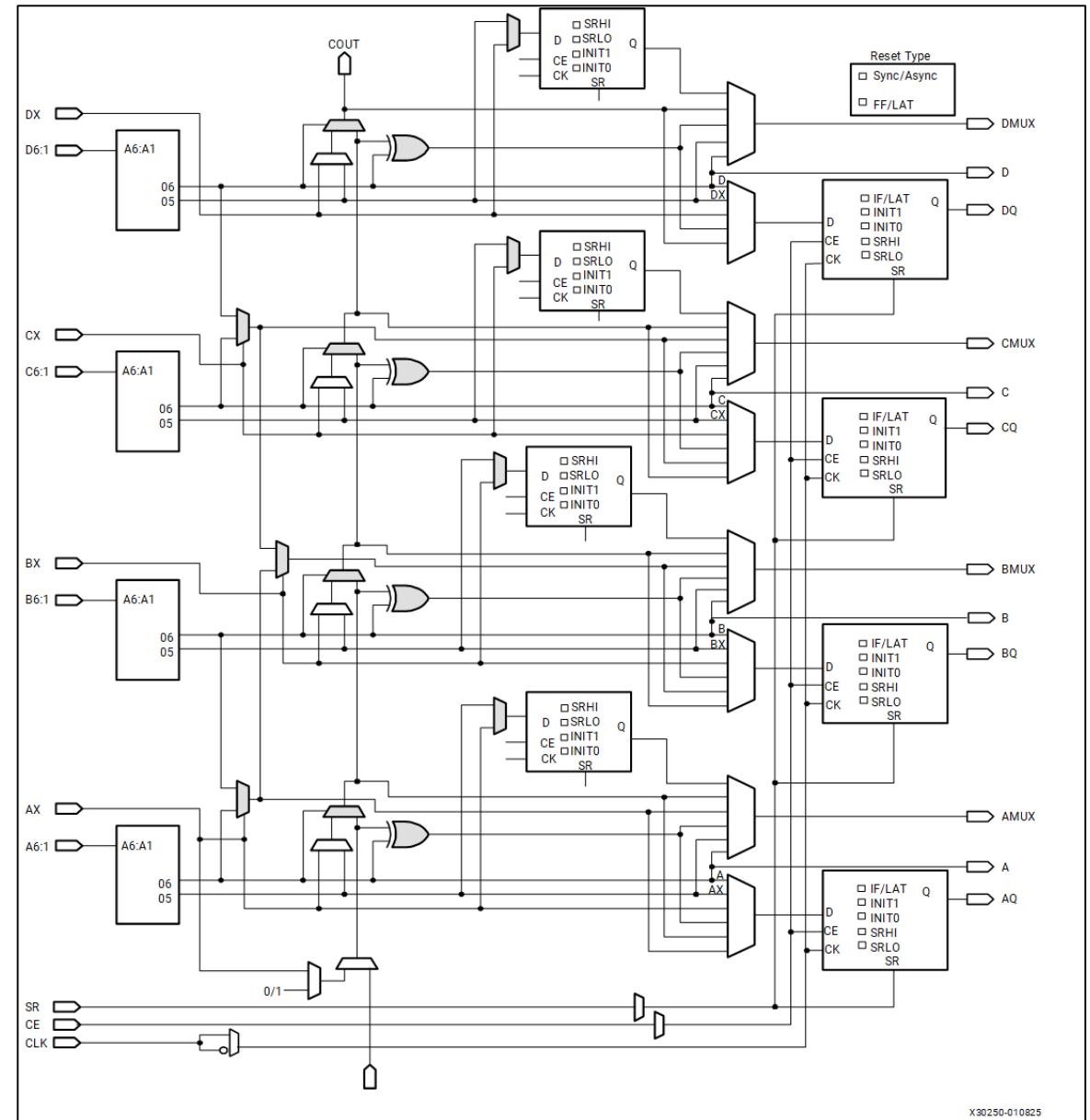
Diagram of SLICEL

- it can be programmed as a shift register (SRL16/SRL32) or as a regular D flip-flop
- When configured as a shift register, this flip-flop supports variable-length shift functionality using the LUT for configuration.
- It allows the designer to choose between registered (synchronous) and combinational (asynchronous) output per logic resource.

SLICEM



SLICEL



Features

- Real 6-input look-up table (LUT) technology
- Dual LUT5 (5-input LUT) option
- Distributed Memory and Shift Register Logic capability
- Dedicated high-speed carry logic for arithmetic functions
- Wide multiplexers for efficient utilization